

SMART INDIA HACKATHON · PS 26146 · NTRO

BEANS

Bitcoin Encryption, Analysis & Network Security

Technical report: implementation, features and measured accuracy

0.981

fused-risk PR-AUC, mean
of 6 synthetic datasets

98.4 %

of criminal entities alerted

0.799

illicit F1 on real Bitcoin
data (Elliptic)

0.98 M

rows loaded + scored in
286 s

Release v1.1 (commit 6a74af1) · generated 2026-09-27 · runs fully offline

All figures in this report are generated from result files in the repository (`scripts/build_report.py`).

1. Summary

BEANS ingests bulk Bitcoin transaction and network-relay observations (CSV, JSON or XML), enriches every relaying IP offline with country and ASN type, and links IPs, transactions and wallets in one graph. Five engines score it (entity clustering, anomaly detection, transaction-shape classification, risk propagation from known illicit seed wallets, and a graph neural network); a calibrated fusion model turns them into a ranked, explained alert list. Every alert then gets a **recommended next step** from fixed, citable rules: draft a freeze request or a Section 94 BNSS notice, prepare an FIU-IND referral, watch the funds, or review a likely false positive. Legal drafts need a second person's approval; evidence is sealed with SHA-256 and an offline RFC 3161 timestamp; critical alerts can be pushed to a SIEM.

0.951

recall of illicit wallets ($P \geq 0.5$), before 0.844

0.989

precision ($P \geq 0.5$), before 0.946

1.000

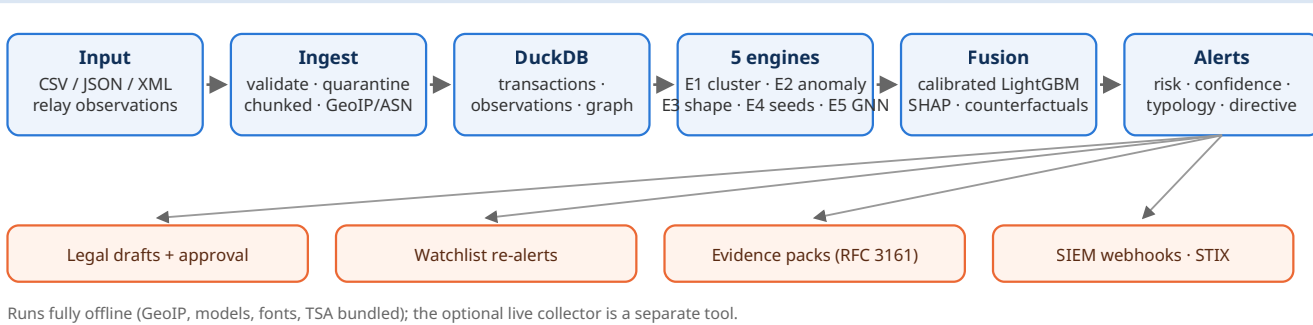
clusters never mix two criminals (all 6 datasets)

0.0037

calibration error (ECE)

How to read the numbers. Synthetic results are means over six independently generated datasets, out-of-fold and grouped by criminal entity (a model is never tested on an entity it trained on). Our generator produced that data, so absolute synthetic numbers are optimistic; the meaningful parts are the before/after comparison, the ablations and the real-data validation on Elliptic (section 5), where BEANS matches the strongest published baseline rather than beating it.

2. System overview



Layer	What it does
Ingest	Streams CSV / JSON / XML in 50,000-row chunks; column mapping for unfamiliar exports; invalid rows quarantined with the reason, never filled with fake values; the earliest relay of a transaction wins across files (first-spy); optional wallet-software fields (version, nLockTime, RBF).
Enrichment	Offline DB-IP country and ASN databases; ASN typed as residential / datacenter / VPN / Tor exit / bulletproof hosting.
Storage	Embedded DuckDB file; investigator state (cases, verdicts, watchlist, approvals, audit trail) survives re-scoring.
Interfaces	FastAPI REST API; React dashboard (served pre-built, no Node.js needed); CLI; Docker image that passes a test with networking disabled.

3. Engines

Engine	Method	Measured (v1.1, 6-dataset mean)
E1 entity clustering	Common-input ownership (CoinJoins excluded) · change heuristics with guards (never a later-swept deposit address, never the tiny peel payment, never spent by different wallet software) · peel-chain change · self-split rule (≥ 7 near-identical fresh parts from a non-hub owner)	homogeneity 1.000 · completeness 0.744 (was 0.580)
E2 anomaly	Isolation Forest on behaviour + network features	used as a fusion input
E3 transaction shape	LightGBM on 31 structural features: peel, CoinJoin, fan-out, fan-in, round trip	macro-F1 0.997
E4 seed propagation	Personalised PageRank (both directions), decayed haircut taint with exchanges as taint sinks, hop distances incl. direction-agnostic (reaches a seed's siblings)	86 % of hidden wallets of seeded actors reached; 9.7 % of legitimate wallets (was 16.9 %)
E5 graph neural network	SIGN-style: 15 wallet signals averaged over 1- and 2-hop neighbours along and against the money flow, service hubs cut out; fusion is the readout; CPU-only, no deep-learning framework	when added: PR-AUC 0.954 → 0.966, recall 0.905 → 0.952
Fusion	LightGBM, isotonic calibration repeated over 3 fit/calibration splits per fold and averaged, out-of-fold by entity; money-context features (funding shape, structuring just below round amounts, deposit-like destinations); typology model with a reference corpus of 253 extra operations, pooled per cluster	PR-AUC 0.981 · ECE 0.0037
Explanations	SHAP reasons in plain English; counterfactuals ("without Tor/VPN relaying the risk would fall from 100 to 17"), flagging single-signal vs corroborated alerts	on every alert (Figure 4)

4. Accuracy on synthetic data

Protocol. `scripts/evaluate_seeds.py --seeds 42 7 123 2024 99 555` generates six independent datasets (~4,000 transactions, ~13,500 wallets, ~30 criminal entities each, only ~30 % of entities revealed as seeds) and runs the full pipeline on each in its own database. "Before" is the code at the start of the accuracy work; "v1.1" is this release (v1.0 plus repeated calibration). Metrics are out-of-fold and grouped by entity.

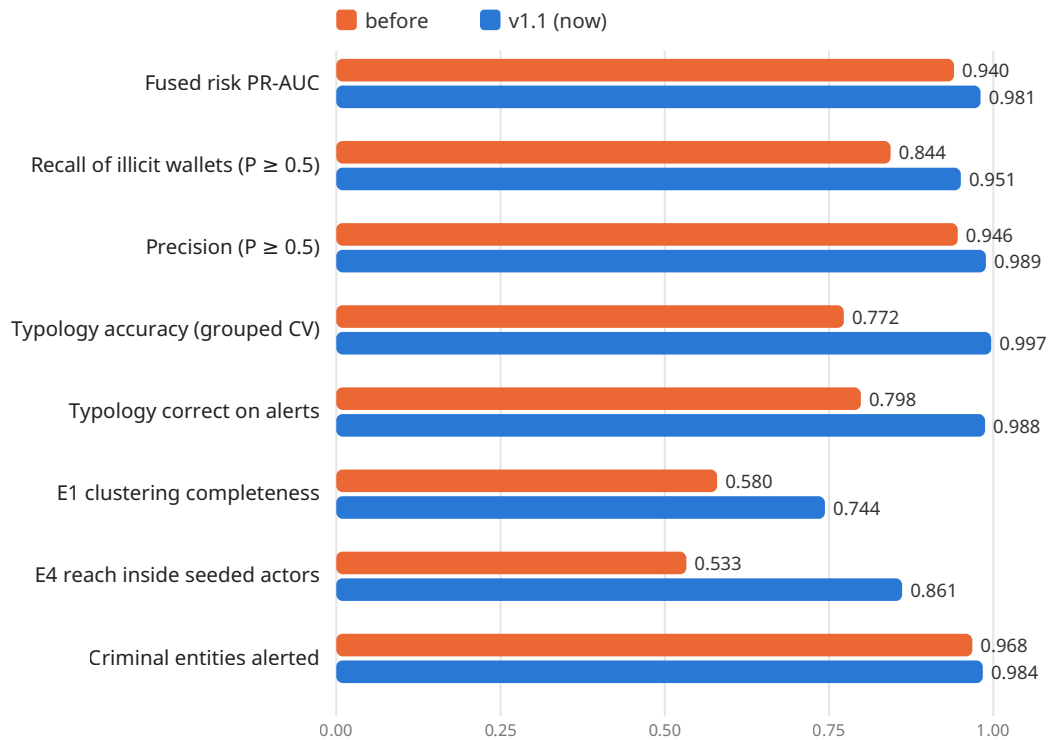


Figure 1. Before vs v1.1, mean of six datasets (values on each bar; the table below has the spread).

Metric (mean of 6 datasets)	Before	v1.1 ($\pm$ std)	
PR-AUC (fused risk)	0.940	0.981 $\pm$ 0.004	▲
ROC-AUC	0.985	0.992 $\pm$ 0.005	▲
Recall at $P \geq 0.5$	0.844	0.951 $\pm$ 0.009	▲
Precision at $P \geq 0.5$	0.946	0.989 $\pm$ 0.002	▲
Calibration error (ECE, lower is better)	0.008	0.004 $\pm$ 0.001	▲
Typology accuracy, grouped CV	0.772	0.997 $\pm$ 0.006	▲
Typology correct on alerts	0.798	0.988 $\pm$ 0.023	▲
E3 transaction-shape macro-F1	0.997	0.997 $\pm$ 0.002	
E1 completeness (illicit)	0.580	0.744 $\pm$ 0.024	▲
E1 homogeneity (illicit)	0.999	1.000 $\pm$ 0.000	▲
E4 hidden wallets reached, seeded actors	0.533	0.861 $\pm$ 0.058	▲
E4 legitimate wallets reached (lower is better)	0.169	0.097 $\pm$ 0.029	▲
Criminal entities with ≥ 1 alert	0.968	0.984 $\pm$ 0.016	▲
Alerts per criminal entity (lower is better)	9.875	5.661 $\pm$ 0.525	▲
Alerts that are illicit	0.943	0.927 $\pm$ 0.016	▼

Repeated calibration (v1.1). In v1.0 each fold calibrated on one 25 % slice of its training actors, only a handful of criminal operations, so the isotonic curve had coarse steps: a step holding one illicit and one licit wallet scores exactly 0.5, and on two datasets 12 of the 15 alerts at exactly 0.5 were innocent. Averaging three fit/calibration splits per fold raised alert precision from 0.891 (v1.0) to **0.927**, better on every one of the six datasets, PR-AUC from 0.967 to 0.981, with criminal entities alerted unchanged (0.984 → 0.984). The Elliptic benchmark (section 5) calibrates on a held-out time window, so this change is not measured there. Alert precision is still below the start (0.943) because each criminal now takes ~5.7 alerts instead of ~9.9: the same few false alerts weigh more in a list half as long.

What changed, stage by stage (each a full 6-dataset run, [docs/results/stages/](#))

Stage	PR-AUC	Recall	Typology	E1 compl.	E4 seeded	Entities
start	0.940	0.844	0.772	0.580	0.533	0.968
+ money-context features	0.953	0.891	0.801	0.580	n/a	0.968
+ clustering fix, self-split, E4 fixes	0.954	0.905	0.831	0.744	0.861	0.962
+ E5 graph neural network	0.966	0.952	0.845	0.744	0.861	0.984
+ typology corpus (v1.0)	0.966	0.952	0.997	0.744	0.861	0.984
+ repeated calibration	0.981	0.951	0.997	0.744	0.861	0.984
v1.1 (release)	0.981	0.951	0.997	0.744	0.861	0.984

- **Money-context features** (label-free): uniformity of the funding transaction's outputs, amounts just below a power of ten (structuring), hub payers, deposit-like destinations.
- **Clustering fix + self-split rule; E4 taint sinks and sibling reach:** an old change-heuristic mistake bridged whole exchanges into criminal clusters; false E4 reach fell 16.9 % → 9.7 %.
- **E5 graph neural network** and the **typology reference corpus** (253 extra operations). A label-shuffling control on the corpus (2 datasets) dropped typology to 0.48-0.61, so the gain comes from its labels; near-perfect accuracy mainly shows synthetic typologies are separable once enough operations are seen.

Tried and rejected (measured, not guessed)

- Counterparty features: PR-AUC 0.954 vs 0.953 without, dropped as no gain. Wallet-level fingerprint features: recall 0.952 → 0.922, dropped; fingerprints are kept only as a clustering guard.
- Self-split rule with ≥ 5 parts: better completeness, but merged a darknet market with its vendors; ≥ 7 keeps homogeneity at 1.00.

Demo dataset (the one the dashboard ships with)

Metric	Value
Fused PR-AUC (without network-layer features)	0.977 (0.921)
Recall / precision at $P \geq 0.5$	0.953 / 0.993
Alerts · criminal entities alerted	141 · 30 of 30
Alerts that are illicit · top 10	94 % · 100 %
Typology correct on alerts	100 %
E1 homogeneity · completeness	1.00 · 0.79
E4 reach inside seeded actors · legitimate reached	77 % · 8 %

5. Validation on real Bitcoin data (Elliptic)

Elliptic (Weber et al. 2019) has 203,769 real Bitcoin transactions (4,545 labelled illicit, 42,019 licit) and 234,355 payment edges. Its features are anonymised (no addresses, amounts or IPs), so it tests the modelling approach, not the clustering, network layer or action rules. Standard temporal split: train on time steps 1-34, test on 35-49 (1,083 illicit test transactions).

BEANS (this code)	P	R	F1	PR-AUC	Published (Weber et al. 2019)	P	R	F1
BEANS LightGBM + calibration (LF)	0.901	0.672	0.770	0.752	Logistic regression (AF)	0.404	0.593	0.481
BEANS LightGBM + calibration (AF)	0.921	0.706	0.799	0.759	Random forest (LF)	0.803	0.611	0.694
BEANS LightGBM + calibration (AF + graph)	0.909	0.713	0.799	0.760	Random forest (AF)	0.956	0.670	0.788
BEANS LightGBM + calibration (LF + GNN (E5))	0.803	0.721	0.760	0.755	Random forest (AF + GCN embeddings)	0.971	0.675	0.796
BEANS LightGBM + calibration (AF + GNN (E5))	0.853	0.718	0.780	0.756	GCN	0.812	0.512	0.628
Random forest (LF), re-run here	0.854	0.718	0.780	0.772	Skip-GCN	0.812	0.623	0.705
Random forest (AF), re-run here	0.891	0.724	0.799	0.775				

Reading. BEANS reaches illicit F1 **0.799** (precision 0.92, calibration error 0.0175), equal to the strongest published baseline and well above the published graph networks; it does not beat the random forest. E5 does not help here because Elliptic's features already are neighbourhood aggregates. Like every published model, detection collapses after time step 43, when a large dark market closed.

Seed propagation on real data. With 30 % of illicit transactions revealed as seeds, 50 % of the hidden illicit ones lie within 2 hops of a seed versus 5 % of licit ones, and adding the seeds lifts PR-AUC from 0.736 to **0.821** (the combination rule was fixed before testing).

6. Scale

	Value
Rows (10 independent files, one database)	980,803
Transactions · wallets	336,260 · 726,035
Chunked ingest (<code>beans ingest ... --no-score</code>)	96 s · peak 0.88 GB
Scoring, all engines (<code>beans score</code>)	190 s · peak 5.29 GB
End to end	286 s · 3,429 rows/s

The earlier estimate for one million rows was ~16 GB of RAM; chunked ingest and two optimised hot spots (identical results, verified to four decimals) bring it to a normal 16 GB laptop. The demo dataset scores end to end in 28 s.

7. Investigator workflow

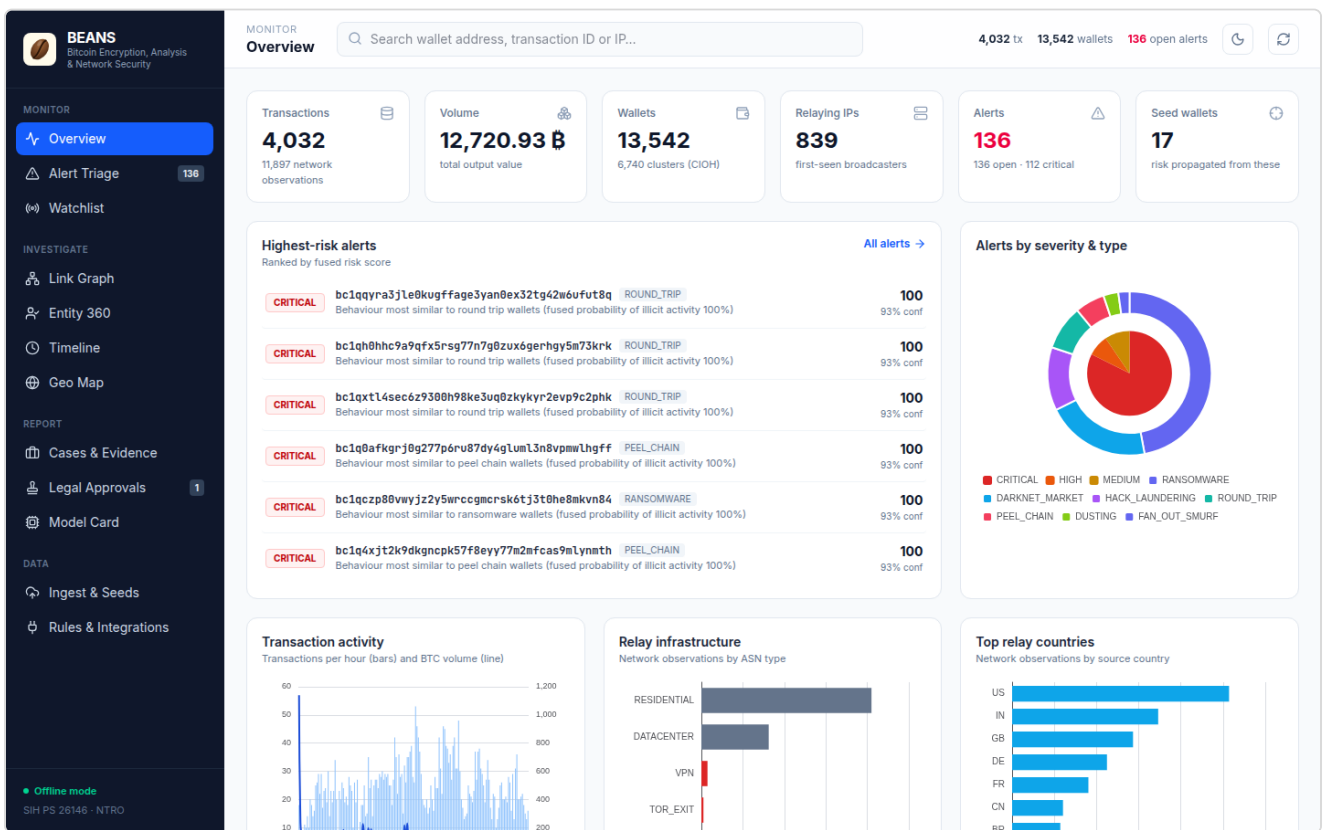


Figure 2. Overview: volume, alerts by severity and typology, relay infrastructure, countries.

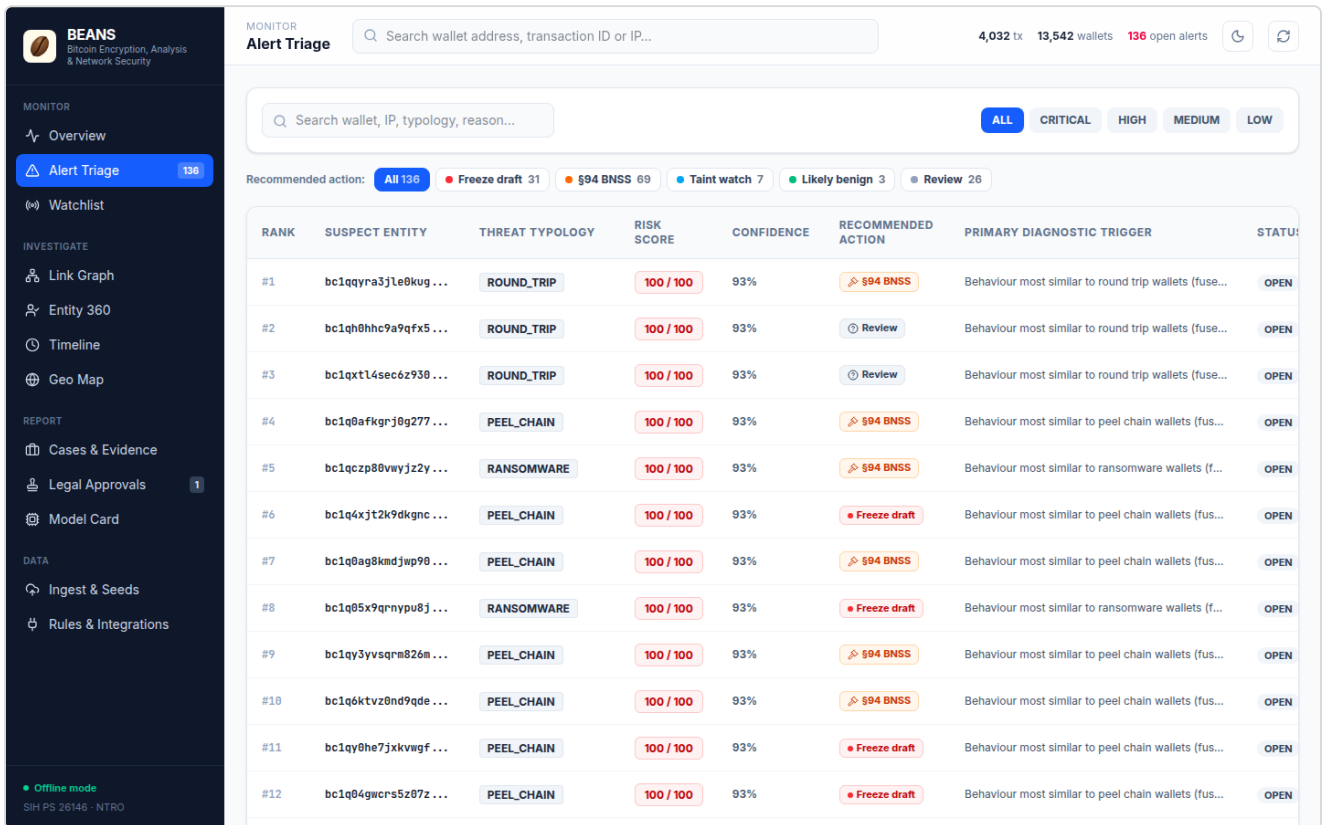


Figure 3. Alert triage: every alert carries a recommended action; filter chips by action; linked-case badges.

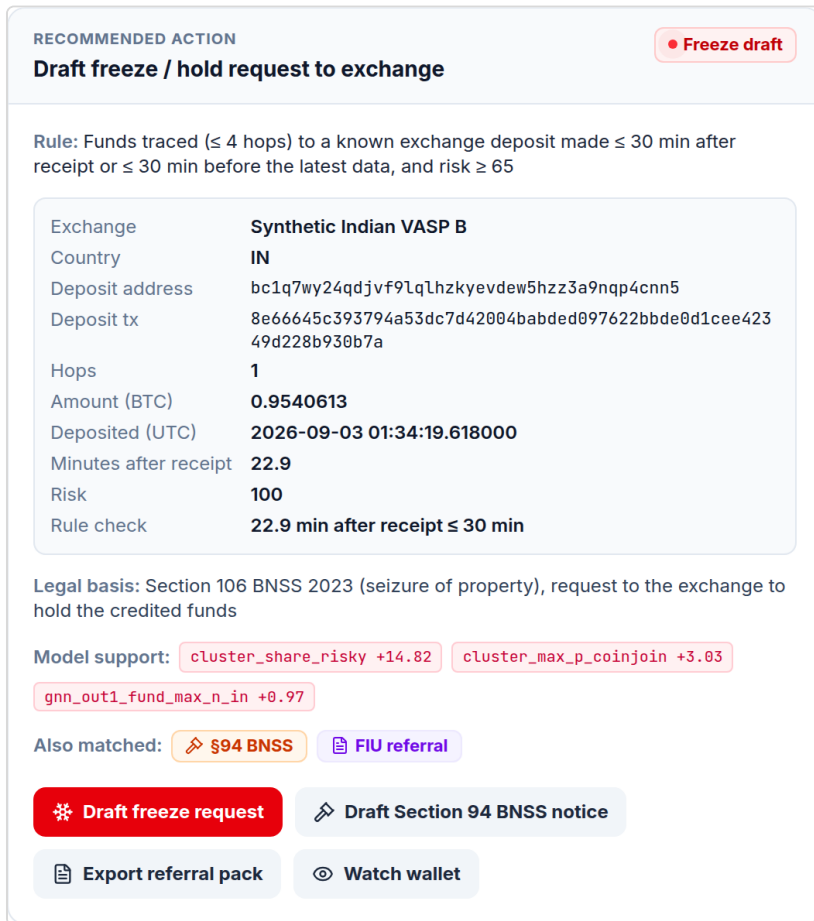


Figure 4. Action card: the directive, the rule it matched with the checked values, legal basis, model support (SHAP) and one-click drafts.



Figure 5. What would change the verdict: each factor neutralised to a typical wallet and re-scored; single-signal vs corroborated alerts.

Action directive rulebook (demo counts)

Directive	Rule	Legal basis	Alerts
Immediate Freeze Draft	Funds traced (≤ 4 hops) to a known exchange deposit made ≤ 30 min after receipt or ≤ 30 min before the latest data, and risk ≥ 65	Section 106 BNSS 2023 (seizure of property), request to the exchange to hold the credited funds	35
Draft Section 94 Bnss	Funds traced (≤ 4 hops) to an exchange that operates in India / is registered with FIU-IND	Section 94 BNSS 2023 (summons to produce documents or other things)	69
Cross Chain Exit	Funds traced (≤ 4 hops) to a known swap service or bridge (e.g. BTC $\rightarrow$ XMR / ETH). A Bitcoin freeze no longer reaches them	Section 94 BNSS 2023 to the swap service (Letter of Request, Section 112 BNSS, if offshore) for the order: payout chain, payout address, IP, account; then trace on the destination chain	0
Fiu Referral Pack	≥ 1 BTC moved, broadcast via Tor / VPN / bulletproof hosting, with a layering pattern (peel chain, CoinJoin or fan-out)	PMLA 2002 intelligence sharing; the pack goes to the IO / FIU-IND liaison, not filed as an STR	0
Passive Taint Monitor	Unspent balance, dormant ≥ 6 h, linked to a seed wallet (taint > 0 or ≤ 4 hops)	No legal step yet. Watch the funds and re-alert when they move	7
Review Likely Benign	The wallet is a known exchange / mining-pool address, or is funded by a mining pool with no seed link and no risky infrastructure	None. Confirm or reject; the verdict feeds model retraining	2
Analyst Review	No directive rule matched. Read the reasons and evidence	None	28

Thresholds (freeze window 30 min, minimum risk 65, trace depth 4 hops, layering ≥ 1 BTC) live in `beans/config.py` and can be overridden without code changes. Exchange deposits come from an attribution list (`known_entities`), which is never a model feature.

Freeze / hold request (draft)

A-W-35eefdf4d · bc1q4xjt2k9dkgncpk...

×

BEANS fills in the blockchain facts. Fields left empty stay as visible blanks for the Investigating Officer. IO details are remembered in this browser only.

Language

Englishहिंदी

FIR / case no.

FIR date

Offences (sections)

IO name

IO rank

Police station

District, State

Reply within (days)

Exchange's nodal-officer address

Redraft (files a new request)

Evidence seal

SHA-256

PENDING SUPERVISOR APPROVAL (request #2, drafted by local). Not for issue.

REQUEST TO FREEZE / HOLD CREDITED FUNDS (SECTION 106,
BHARATIYA NAGARIK SURAKSHA SANHITA, 2023)

FIR / Case No. [] dated [], P.S. [], []

To,
The Principal Officer / Nodal Officer (Law Enforcement Requests)
Synthetic Indian VASP B
[]

Subject: Immediate hold on funds credited to cryptocurrency deposit address(es): FIR / Case No.
[] dated [], P.S. [], []

During investigation of the above case, registered for offences under [], proceeds suspected to be connected with the offence were traced on the Bitcoin blockchain to the following deposit address(es) of your exchange. The first deposit reached your exchange **22.9 minutes** after the flagged wallet received the funds, so the funds are likely still held by you:

1. bc1q7wy24djvf9lqlhzkyevde5hzz3a9nqp4cnn5

2. bc1qfnt37an35093u0ac4p15tldks433qv906l07lup9jwe0nh7kn4ka3nsbw8

3. bc1qqpske5t43p7fvzv5ryfa39vrajt93xxsv8uqyp6

In exercise of the powers under Section 106 of the Bharatiya Nagarik Suraksha Sanhita, 2023, you are requested to **immediately freeze / place on hold** all balances (crypto and INR) of the account(s) to which these addresses are assigned, to block withdrawals, and to confirm the action and the balances held to the undersigned. A requisition for KYC and account records under Section 94 BNSS follows separately.

(Signature)
[],
Investigating Officer
P.S. [], []
Date: []

Annex A: blockchain evidence

Flagged wallet bc1q4xjt2k9dkgncpk57f8eyy77m2mfcas9lmnmth · alert A-W-35eefdf4d ·

BEANS

Bitcoin Encryption, Analysis & Network Security

MONITOR

Overview

Alert Triage129

Watchlist2

INVESTIGATE

Link Graph

Entity 360

Timeline

Geo Map

REPORT

Cases & Evidence

Legal Approvals

Model Card

DATA

Ingest & Seeds

Rules & Integrations

Offline mode

SIH PS 26146 - NTR0

REPORT

Legal Approvals

Search wallet address, transaction ID or IP...

4,034 tx

13,543 wallets

129 open alerts

Test SUPERVISOR SUPERVISOR

Legal drafts

Section 94 and freeze drafts need a supervisor's approval, by someone other than the drafter, before a final copy can be issued.

Pending

Approved

Rejected

All

#2 · Section 94 BNSS notice

APPROVED

Synthetic Indian VASP B · wallet bc1qy8he7jkkvw...

drafted by ana · 2026-09-26 14:37 · approved by sup

#1 · Section 94 BNSS notice

APPROVED

Synthetic Indian VASP B · wallet bc1qy8he7jkkvw...

drafted by ana · 2026-09-26 14:36 · approved by sup

Request #2 · Section 94 BNSS notice

APPROVED

evidence SHA-256 c47fc392b4aa84da...

Approved PDF

APPROVED FOR ISSUE by supervisor sup on 2026-09-26 14:37:41 UTC. Request #2, drafted by ana. The Investigating Officer must still sign before issue.

NOTICE UNDER SECTION 94 OF THE BHARATIYA NAGARIK SURAKSHA SANHITA, 2023

FIR / Case No. 45/2026 dated [_____] , P.S. [_____] , [_____]

To,

The Principal Officer / Nodal Officer (Law Enforcement Requests)

Synthetic Indian VASP B

[_____]

Subject: Production of KYC and account records relating to cryptocurrency deposit address(es):

FIR / Case No. 45/2026 dated [_____] , P.S. [_____] , [_____]

Whereas the investigation of the above case, registered for offences under [_____] , is in progress, and it has come to notice that proceeds connected with the case were deposited to the following address(es) controlled by your exchange:

1. bc1q8s9y4xs2ymxm4zyj3r5fawc8hswlkhmncz3ucg

2. bc1qasckvkuscshn6tu6qe5ndncayuzngmt36jqn9hg

And whereas the production of the documents and records listed below is necessary for the purpose of the said investigation, you are hereby required under Section 94 of the Bharatiya Nagarik Suraksha Sanhita, 2023 to produce them, in certified electronic form with a certificate under Section 63 of the Bharatiya Sakshya Adhiniyam, 2023, within **7 days** of receipt of this notice:

1. KYC documents of the account holder(s) to whom each address above is assigned (name, PAN, photograph, mobile number, e-mail, address proof) and the account creation date.

2. Complete ledger of deposits, trades and withdrawals (crypto and INR) of those accounts from 2026-09-04 to date.

3. Login, IP address and device logs with timestamps for the same period.

BEANS - SIH PS 26146 - runs fully offline - IP geolocation by DB-IP (CC BY 4.0) - the loaded dataset is synthetic (generated with ground truth)

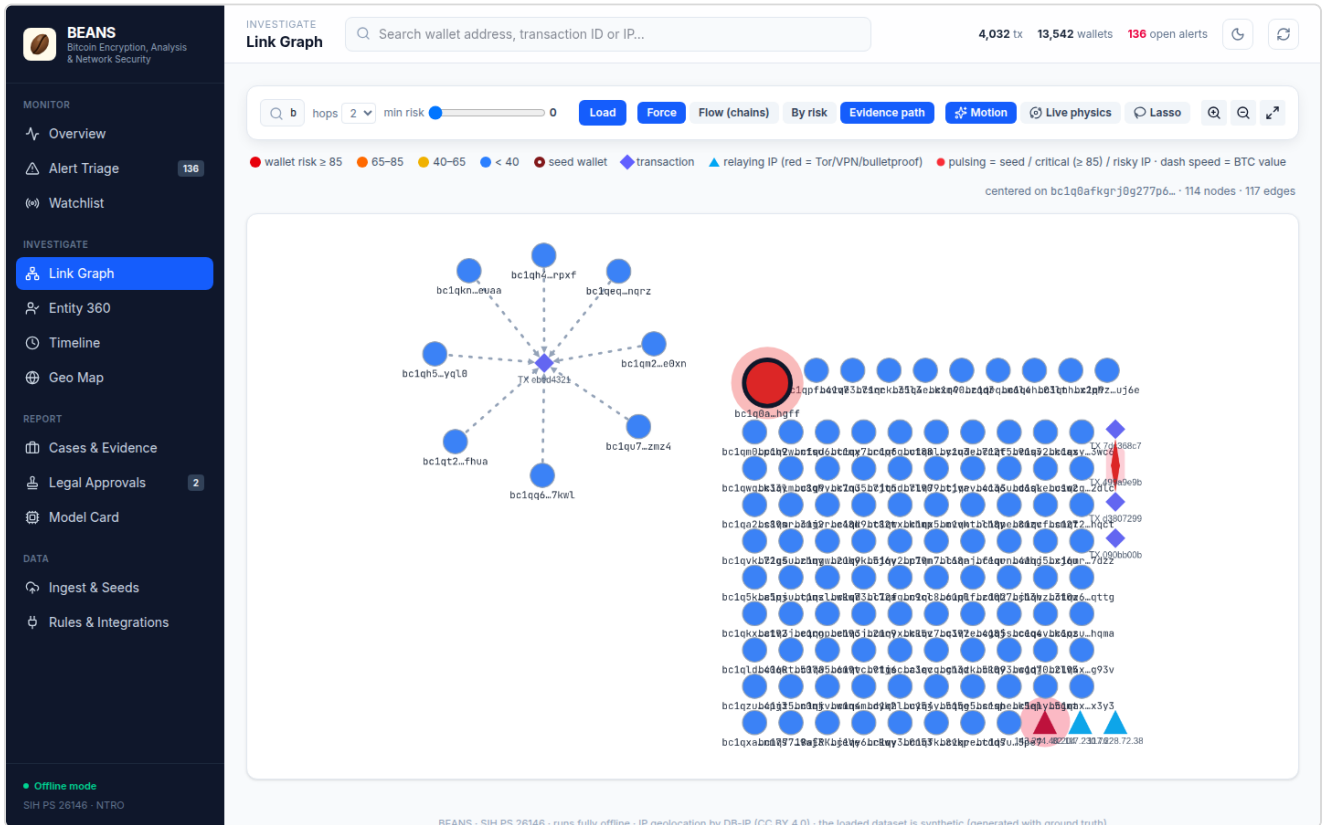


Figure 8. Link graph: money-flow particles (speed = BTC), pulsing halos on seeds / critical wallets / risky IPs, lasso selection into a case.

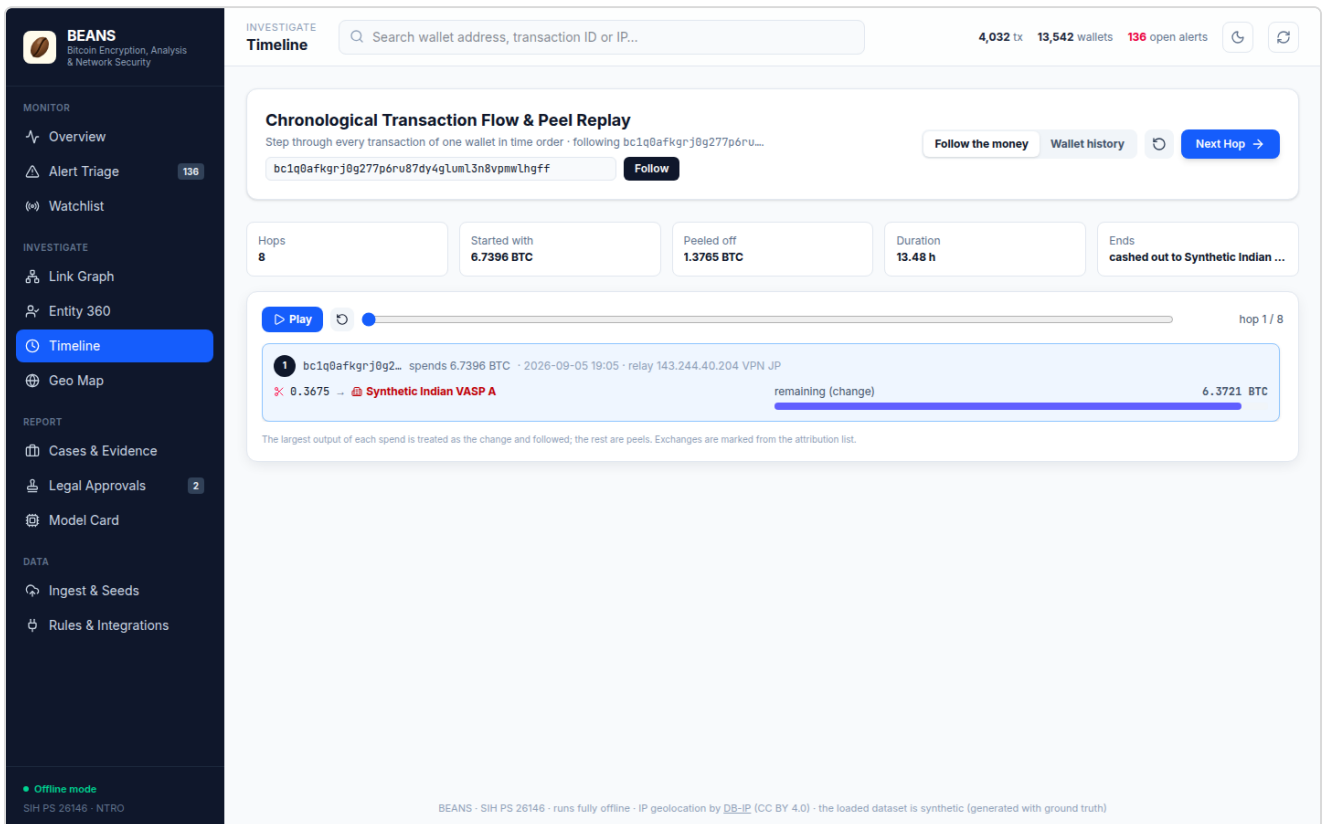


Figure 9. Follow the money: hop-by-hop peel-chain replay with peeled amounts and the exchanges they reached.

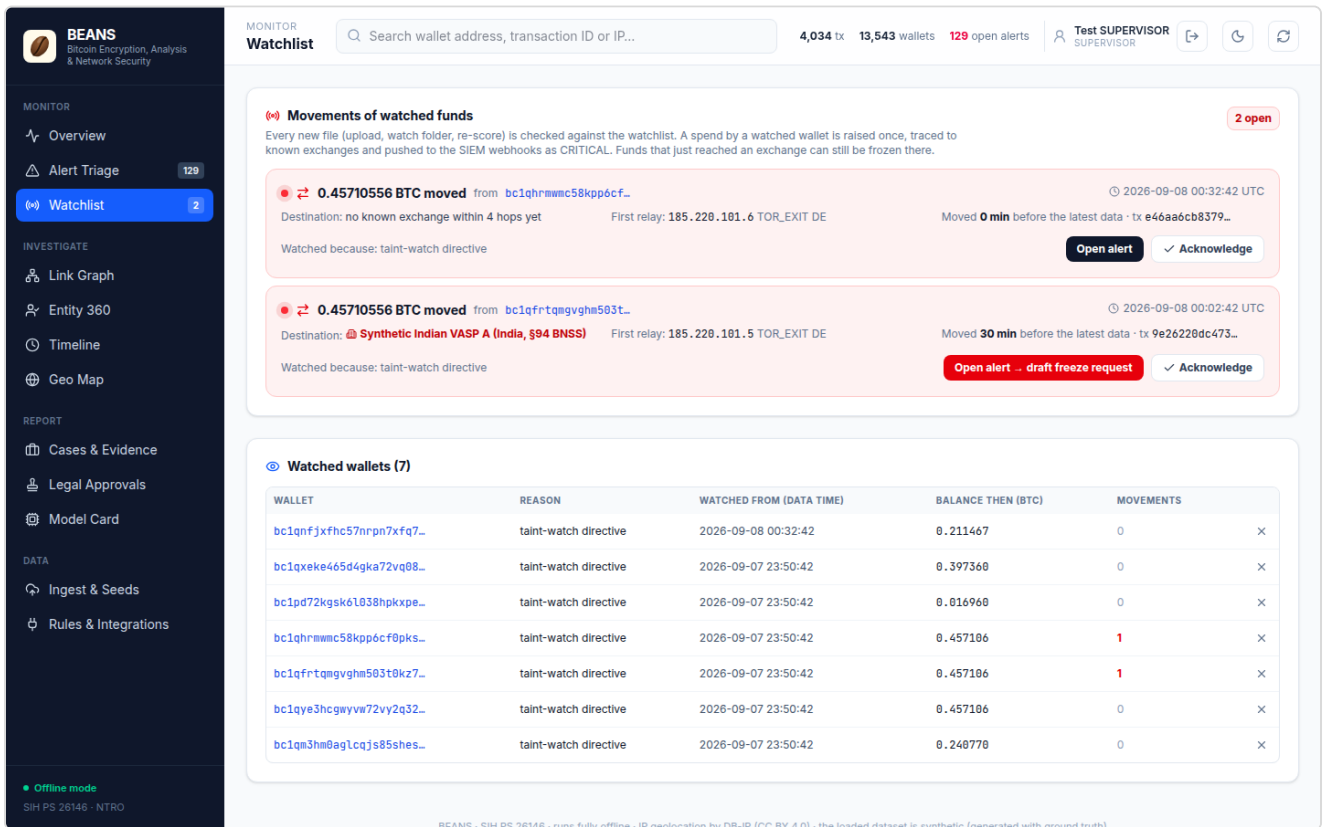


Figure 10. Watchlist: a watched wallet moved funds; one reached an Indian exchange within the freeze window.

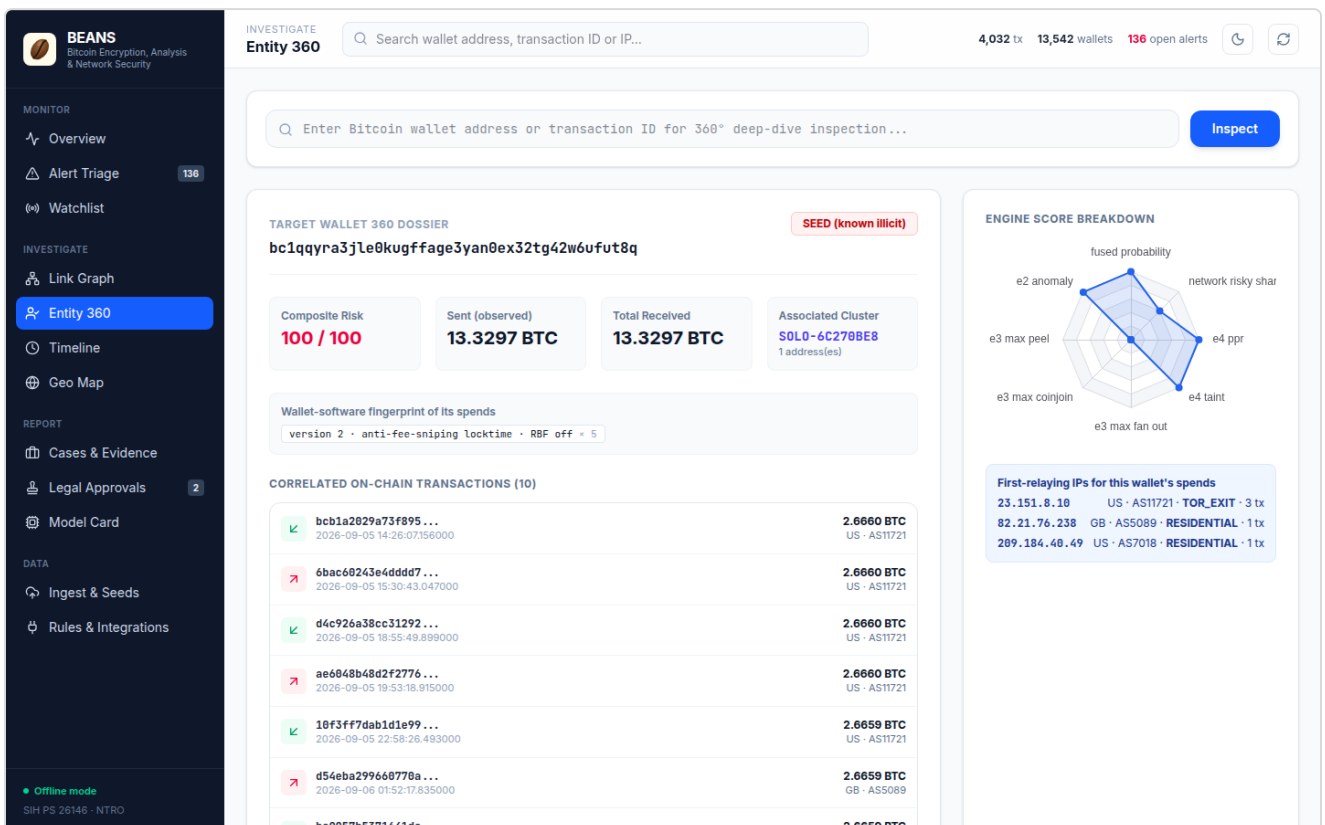


Figure 11. Entity 360: wallet profile, cluster, relaying IPs, wallet-software fingerprint and full ledger.

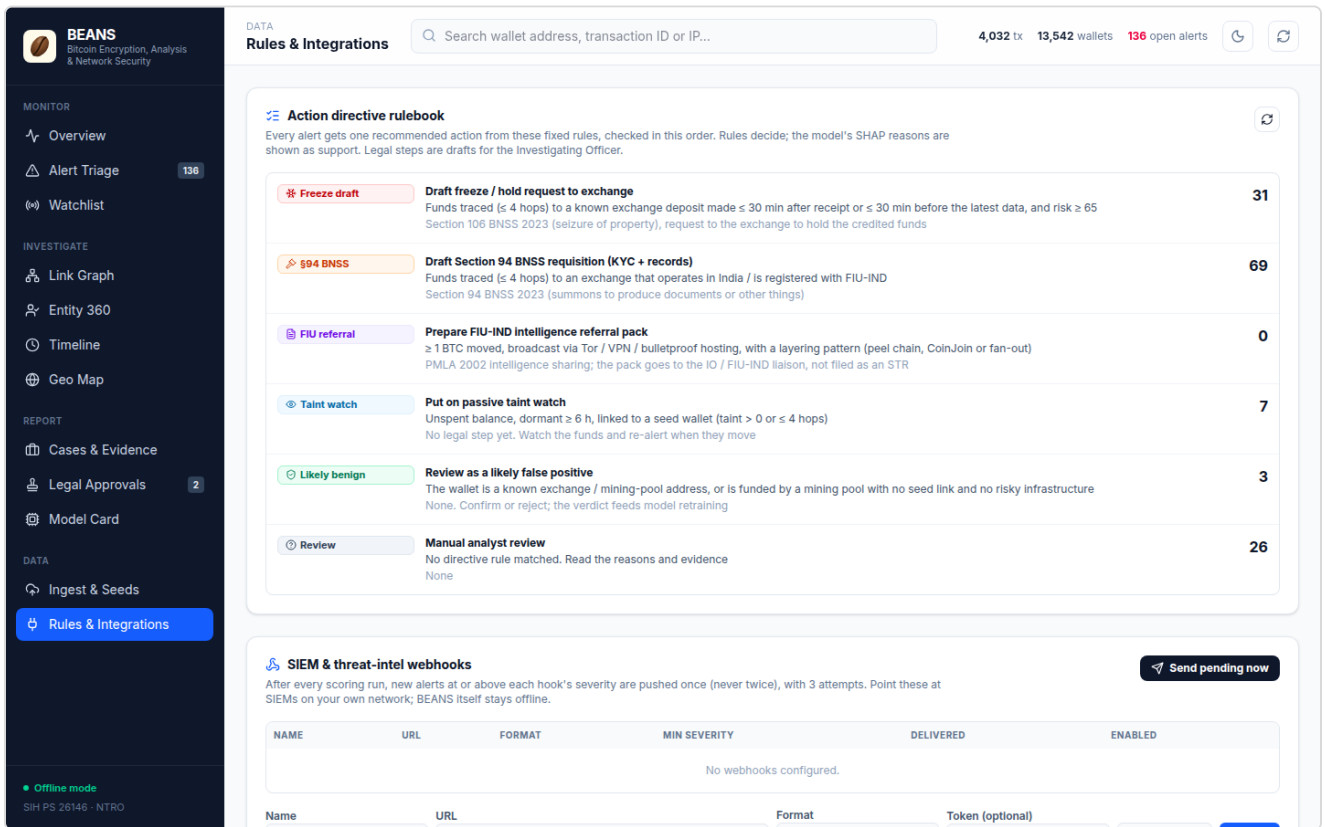


Figure 12. Rules & integrations: directive rulebook, SIEM webhooks, attribution list, evidence timestamping.

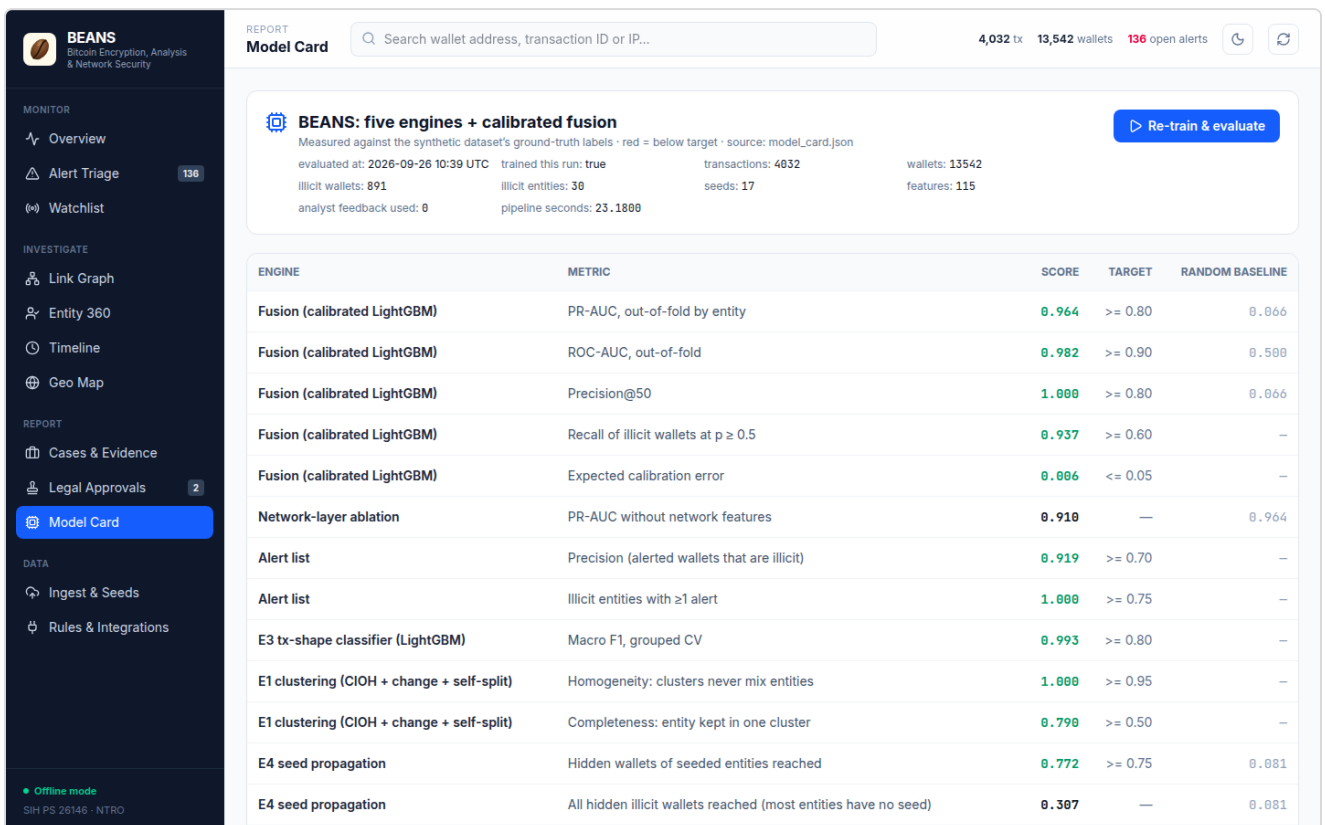


Figure 13. Model card: every metric measured on the loaded data, with targets and baselines, plus the Elliptic validation.

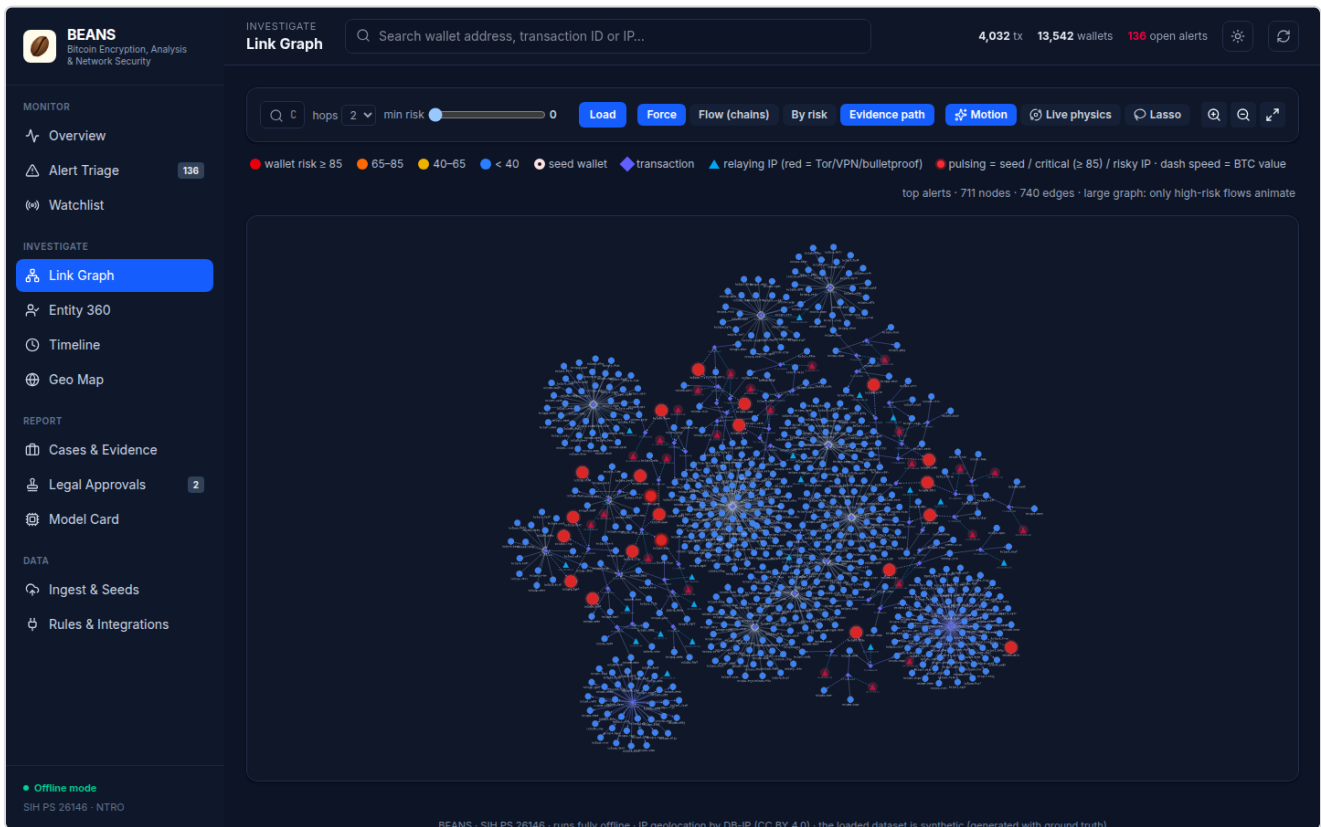


Figure 14. Dark theme (all pages).

The screenshot shows the BEANS login form. The form is titled 'BEANS Bitcoin forensics · sign in'. It includes a 'Username' field, a 'Password' field, and a 'Sign in' button. The 'Sign in' button is highlighted in blue. Below the sign-in button, there is a note: 'Accounts are created by an administrator (beans user add). All actions are recorded in the audit trail under your name.'

Figure 15. Login (switches on once the first user exists).

8. Legal output and evidence

- **Section 94 BNSS requisition** (KYC, ledger, login/IP logs, linked bank/UPI, withdrawal destinations; certificate under Section 63 BSA) and **freeze / hold request** (Section 106 BNSS) in **English or Hindi**; offshore exchanges get a Letter-of-Request (Section 112) note; the freeze letter states its actual trigger (fast cash-out, or a deposit within the window before the latest data).
- **Four-eyes approval**: every draft is filed pending; only a supervisor other than the drafter can approve (or reject with a comment); the approval banner is printed on the document.
- **FIU-IND intelligence referral pack** and **case evidence packs** (PDF / HTML / JSON) with the money-trail and seed-path diagrams, counterfactuals, source-file hashes and the audit trail.
- **Sealing**: SHA-256 of the evidence plus an RFC 3161 token from a local, offline Time-Stamping Authority; anyone can re-verify with `openssl ts -verify`.

9. Monitoring, integrations and security

- **Watch folder** (`beans watch`) and **watchlist**: taint-watch wallets are watched automatically; any later spend raises one CRITICAL movement event, traced to exchanges.
- **SIEM webhooks**: generic JSON (Wazuh), Splunk HEC, Elasticsearch bulk, STIX 2.1 (MISP / OpenCTI); delivered once per alert and hook, retried, logged.
- **Users and roles**: VIEWER / ANALYST / SUPERVISOR / ADMIN; salted PBKDF2 passwords, HttpOnly sessions, lockout after repeated failures; every action in the audit trail under the user's name. No default password: single-user mode until the first user is created.
- **Cross-case links**: an alert shows when its wallet or its entity cluster is already a suspect in an open case.
- **Optional live collector** (`beans collect`): a separate tool that records transaction announcements per peer into BEANS CSVs; checked live (1,030 rows in 90 s, every txid matched the announced id).

10. Added in v1.1

Live data, with real seeds

`beans collect` recorded 12,436 live mainnet transactions (43,761 relay observations from 13 peers, 4,720 wallets). Input addresses are now read from the signature data, so the share of transactions with no input address fell from 20.9 % to **2.5 %** and common-input clustering works on live data; input amounts of confirmed coins come from an optional resolver (own Bitcoin Core node via `gettxout`, or Esplora) that gives up within a minute when unreachable. `beans ofac-seeds` loads the 532 Bitcoin addresses on the US Treasury OFAC SDN list as citable seeds (publish date and file hash recorded). Result on this capture: **0 alerts**, highest fused probability 0.394 (threshold 0.4), and 0 sanctioned addresses moved. That is the correct outcome for ordinary traffic: models trained on synthetic data do not accuse anyone without seeds or analyst verdicts.

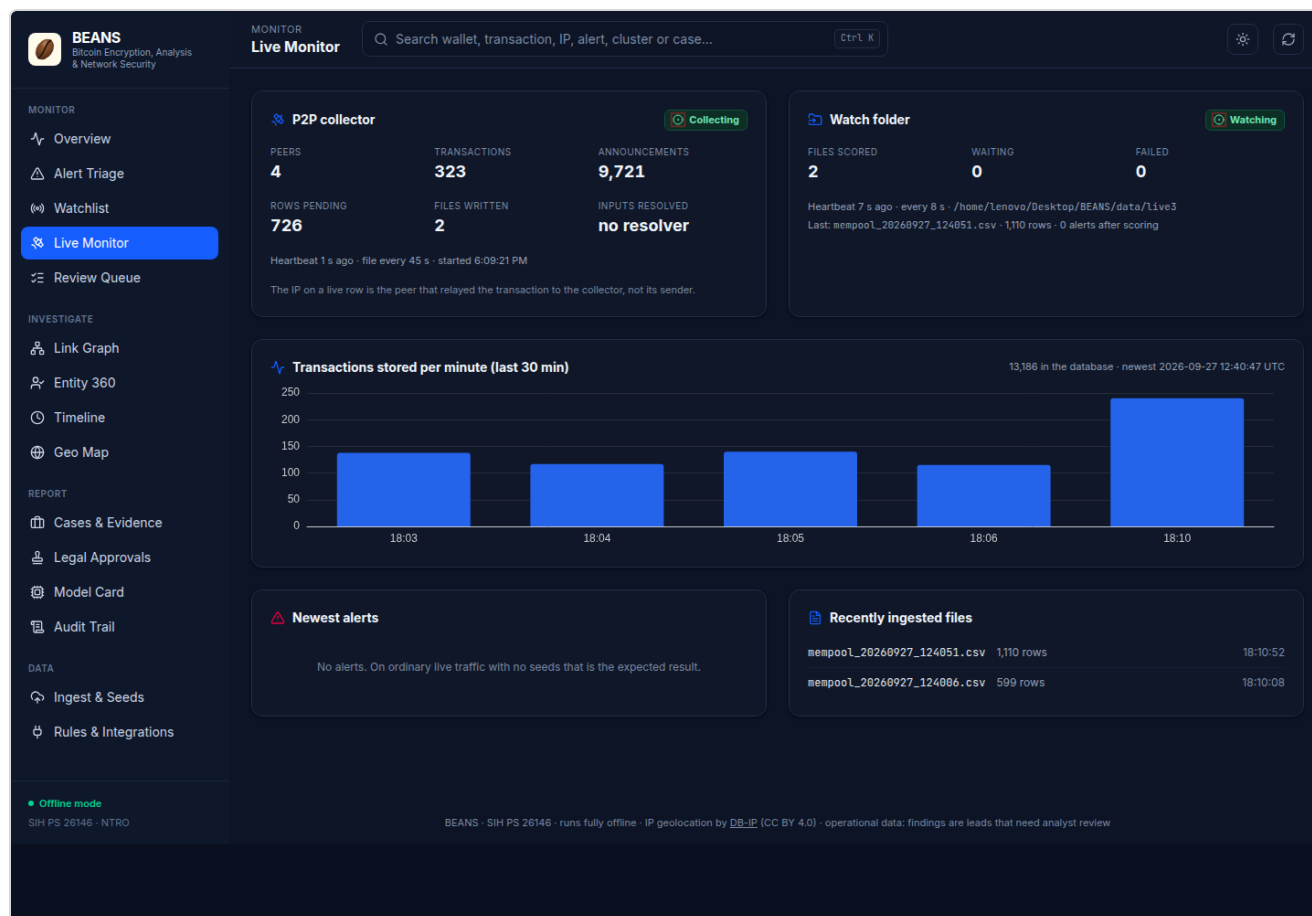


Figure 16. Live Monitor: collector and watch-folder heartbeats, transactions stored per minute, newest alerts and ingested files (live mainnet data).

Review queue (active learning)

Open alerts the model is least sure about (probability near 0.5, low confidence) and wallets just under the alert threshold, where a missed criminal would hide. Each verdict is training feedback; "Retrain now" re-scores with it.

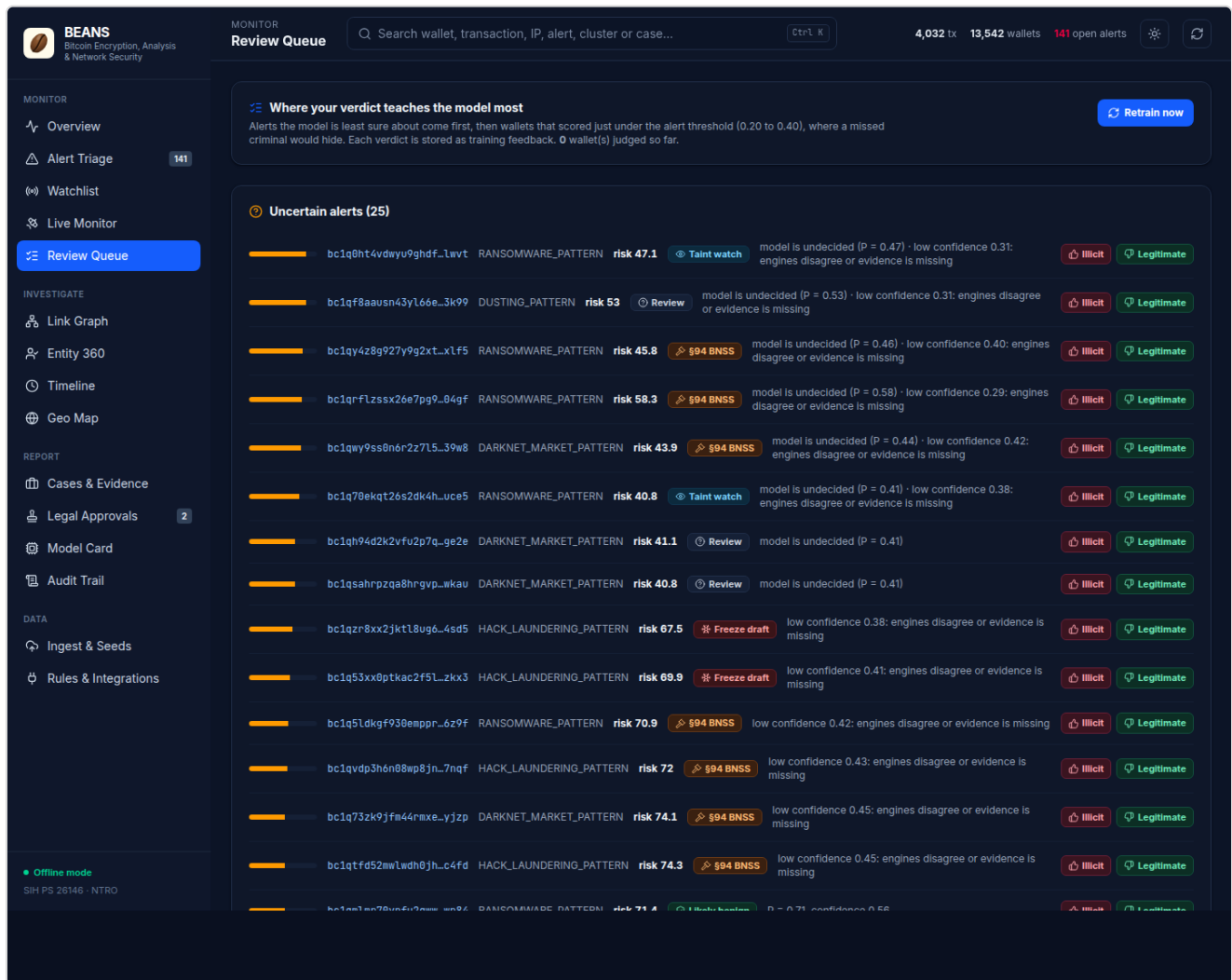


Figure 17. Review queue: uncertainty bar, reason for review, one-click verdicts.

Case summaries from a local model, fact-checked

BEANS builds a numbered fact sheet from its own findings; a local model (Ollama, nothing leaves the machine) writes a short paragraph citing fact ids after every sentence. A sentence is kept only if its facts exist, every number in it appears in them, most of its words come from them and it does not speculate; otherwise it is removed, and if too little survives the fact sheet itself is shown. On a regression sample the check kept all 4 correct sentences and removed all 3 faulty ones (invented number, wrong citation, speculation). Trial on three alerts:

Model	Alert	Seconds	Kept	Removed	Outcome
llama3.2 (3B)	PEEL_CHAIN	32	3	1	model text
llama3.2 (3B)	DARKNET_MARKET	28	2	2	model text
llama3.2 (3B)	ROUND_TRIP	31	0	4	fact-check failed, template
mistral (7B)	PEEL_CHAIN	120	0	0	timeout, template
mistral (7B)	DARKNET_MARKET	120	0	0	timeout, template
mistral (7B)	ROUND_TRIP	90	6	0	model text

The 3B model is fast enough but its text can still join two true facts into a sequence the evidence does not state; the 7B model writes better but needs a longer timeout on a CPU. The summary is labelled as a reading aid; the evidence pack stays the record.

Also new

- **Tamper-evident audit trail:** every entry carries the SHA-256 of the previous one; the Audit Trail page (supervisors) verifies the chain; refused requests are recorded; tighter roles for data-wiping, seed removal and model re-evaluation.
- **Cross-chain exits:** attribution entries of type SWAP / BRIDGE end the trace with a `CROSS_CHAIN_EXIT` directive (request the service's records, trace the other chain), because a Bitcoin freeze no longer reaches the funds.
- **Global search** (Ctrl+K) over wallets, transactions, IPs, alerts, clusters and cases; money-trail diagram in Section 94 / freeze drafts; every ingest path (upload, `beans ingest`, `beans watch`) now records the file hash for case packs.

The screenshot shows the BEANS Audit Trail interface. The left sidebar contains navigation links for MONITOR (Overview, Alert Triage, Watchlist, Live Monitor, Review Queue), INVESTIGATE (Link Graph, Entity 360, Timeline, Geo Map), REPORT (Cases & Evidence, Legal Approvals, Model Card, Audit Trail), and DATA (Offline mode). The main panel displays the Audit Trail with a search bar and a 'Verify chain' button. A green banner at the top of the main panel states: 'Chain intact: 1 entries verified (2 older entries predate the chain). Head c816df7456798ed9...'. Below this, there are filters for User (all) and Action (all), and a 'Filter' button. The audit trail table has the following data:

#	When (UTC)	User	Action	Target	Details
3	2026-09-27 12:13:07	local	NARRATIVE	ALERT: A-W-4f72e7eed2	{ "dropped": 0, "engine": "template", "facts_sha256": "e311600e18e642..." }
2	2026-09-26 18:12:43	local	LEGAL_DRAFT	ALERT: A-W-35eefdf4d4	{ "kind": "freeze", "vasp": "Synthetic Indian VASP B", "request_id": ... }
1	2026-09-26 18:11:00	local	LEGAL_DRAFT	ALERT: A-W-35eefdf4d4	{ "kind": "freeze", "vasp": "Synthetic Indian VASP B", "request_id": ... }

At the bottom of the interface, there is a footer with the text: 'BEANS · SIH PS 26146 · runs fully offline · IP geolocation by DB-IP (CC BY 4.0) · the loaded dataset is synthetic (generated with ground truth)'.

Figure 18. Audit trail with hash-chain verification.

11. Limitations

- Synthetic numbers are optimistic; Elliptic validates the modelling approach only (no addresses, amounts or IPs).
- Typology near 100 % reflects separable synthetic typologies; expect less on real cases.
- Freeze and Section 94 directives on real data depend on a real exchange attribution list.
- Legal templates need review by a lawyer; the Hindi text by a native legal reviewer (the Hindi document says so).
- Without a resolver, the live collector knows input amounts only for coins created by transactions it saw; addresses are read from signatures. The live "first relay" IP is the collector's peer, not the sender.
- Case summaries from a small local model can link true facts wrongly; the fact check removes invented numbers, wrong citations and speculation, not every wrong connection.
- Cross-chain exits need a swap / bridge attribution list; no list of real services is bundled.
- The 1M-row figures predate the memory optimisations in v1.1 and were not re-measured.

- The local TSA proves "no later than" by this machine's clock and key; it carries less weight than an accredited public TSA.

Appendix: reproduce

What	Command
Demo (offline)	<code>make install && make demo</code> → <code>http://127.0.0.1:8000</code>
Offline Docker test	<code>make docker-offline-test</code>
6-dataset benchmark	<code>.venv/bin/python scripts/evaluate_seeds.py --seeds 42 7 123 2024 99 555</code>
Real-data validation	<code>beans validate-elliptic --download --doc docs/VALIDATION_ELLIPTIC.md</code>
1M-row benchmark	<code>.venv/bin/python scripts/benchmark_1m.py --files 10</code>
This report	<code>.venv/bin/python scripts/build_report.py</code>
Tests	<code>make test</code>

IP geolocation by DB-IP (CC BY 4.0). Elliptic dataset: Weber et al., "Anti-Money Laundering in Bitcoin", KDD 2019 workshop (arXiv:1908.02591). SIGN: Frasca et al. 2020. All data shown in screenshots is synthetic.